SANTHOSH R

Bengaluru, Karnataka ◆ 8550038177 ◆ santhosh.dev.6704@gmail.com
Portfolio: santhu.site ◆ github.com/Santhu32144 ◆ linkedin.com/in/santhosh-r-64259b3b0

Full Stack Engineer ◆ React + TypeScript ◆ Node.js ◆ Embedded Linux ◆ Industrial IoT & Real-Time Systems

SUMMARY

Full stack engineer, 2.5+ years building software for **industrial automation hardware** — a **React + TypeScript** operator UI and **Node.js/Express** backend shipped on customer-facing power distribution units. Works across the stack and down to the device: real-time telemetry over **Socket.IO**, **JWT** auth with role-based access, **Modbus / CAN** field protocols, and **embedded ARM Linux** — from the Yocto-built OS image to systemd and nginx deployment.

SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, Kotlin, C, C++, SQL
- **Frontend & Mobile:** React 18/19, Redux Toolkit, Zustand, React Router, Tailwind CSS, Recharts, Vite, React Native CLI
- **Backend:** Node.js, Express 5, REST API design, Socket.IO / WebSockets, JWT (access + refresh), RBAC, Swagger/OpenAPI, Winston
- **Python:** FastAPI, Flask, SQLAlchemy, Pydantic, Selenium, Tkinter, pandas / scikit-learn (gradient-boosted models), Docker & PyInstaller packaging
- **Databases & Architecture:** SQLite, Firebase/Firestore + security rules, PostgreSQL (Supabase, migrations, RLS), MySQL, MongoDB; system design, real-time data pipelines, API contract design
- **Embedded Linux:** Yocto Project (Kirkstone), BitBake, devtool, U-Boot, kernel config, device tree, NXP i.MX8M Plus
- **Linux & DevOps:** Ubuntu and embedded ARM Linux, Bash scripting, systemd units, nginx (reverse proxy + static hosting), udev automount, journald, Git, GitLab, Jest, `pkg` (ARM64 binaries), Vercel, Postman
- **Domain:** Industrial IoT and automation, CAN / Modbus / I2C fieldbus, embedded Linux HMI & touch panels, real-time device telemetry, edge-to-cloud data flow

WORK EXPERIENCE

Senior Software Engineer — Full Stack
06/2025 – Present

Delta Electronics India, Bengaluru

- Sole developer of the **PDU Gen-3 platform**: an embedded React touch-panel UI, a **Vite + React + Tailwind** admin configurator, and a Node.js/Express microservice — shipped on customer-facing Power Distribution Units.
- Designed and built the backend from scratch in **TypeScript**: 17 route modules covering measurement, alarms, calibration, event/data logs, configuration, user management and system control, documented with **Swagger/OpenAPI**.
- Implemented **JWT authentication** with short-lived access tokens and httpOnly refresh cookies, plus **role-based access control** where permissions are re-resolved on every refresh so revocation applies instantly to live sessions.
- Built a **real-time energy sampler** that runs independently of the UI — samples power every 5s into a rolling one-hour window and mirrors it to disk, so data collection never pauses on navigation and survives a backend restart.
- Replaced per-page config fetching with a **boot-time bootstrap and API self-test**, surfacing a live "backend disconnected" indicator by reusing the existing Socket.IO signal — no extra heartbeat endpoint needed.
- Automated device deployment with a **Bash self-extracting installer** that ships the packaged **ARM64 Linux binary** and both web builds, provisions **systemd** units and an **nginx** reverse proxy, and configures **udev**-driven USB automount — with versioned OTA updates and rollback backups.
- Added structured **journald** logging, USB import/export and DB backup/restore for field servicing.
- Customised and built the PDU's **embedded Linux OS** with the **Yocto Project** on an **NXP i.MX8M Plus**: **U-Boot** and kernel changes via `devtool` (device tree, boot config, HDMI/UART drivers) plus custom `meta-delta` recipes.
- Fixed a displaced, flickering boot splash: HDMI negotiated 1920×1080 on a 1280×800 panel, and a stale **U-Boot environment** in eMMC survived every reflash — so forced the mode in the kernel command line instead.
- Found kernel config edits were silently dropped: **BitBake** hashed the fragment's path, not its contents, so sstate reused a stale kernel and builds "passed" having rebuilt nothing. Routed it through `SRC_URI` so changes reach the image.
- Restored GPIO on the **FT260** USB-to-I2C/UART bridge, whose shared pins the UART re-claimed at an unpredictable point after boot; with the driver a prebuilt module, shipped a **systemd** guard service as a new Yocto recipe.

Graduate Engineer Trainee — Frontend
08/2024 – 05/2025

Delta Electronics India, Bengaluru

- Built the frontend of a customer-facing **PDU touch panel** in **React and TypeScript**, overhauling a legacy layout into a responsive interface: architected state with **Redux** and built a reusable component library from **Figma** designs, within hardware constraints (fixed resolution, touch-only input, custom on-screen keyboard).

Software Engineer Intern
01/2024 – 07/2024

Delta Electronics India, Bengaluru

- Worked with **CAN, Modbus and I2C** protocols in industrial hardware; gained exposure to **Linux CLI** tooling and **GitLab**, and reworked PDU UI/UX mockups in **Figma**.

PROJECTS

Jourvia — Group Travel Companion App *React Native 0.85, TypeScript, Zustand, Firebase, Kotlin* *Android · v1.0.2*

- Trip planner for groups on the bare **React Native CLI** (New Architecture, no Expo), ~36k lines of TypeScript: routes parsed from a Google Maps link (geocoded, real road distance via OSRM, weather per stop), friends, 1-to-1 chat with voice notes, packing lists, expense splitting that settles up in the fewest payments, and a hold-to-trigger SOS.
- Built a **live trip mode**: GPS auto check-in at each stop (300 m radius, 2-min dwell, auto-skip, ETAs) running in an **Android foreground service**, plus live location sharing and arrival alerts for companions; check-ins are saved on-device first, then synced.
- Wrote native **Kotlin** modules — 5 home-screen widgets and a voice recorder — and a type-safe **i18n engine** covering English and 9 Indian languages with plurals and fallback.
- Owner / editor / member access in deny-by-default **Firestore security rules**; stale-while-revalidate caching over **Zustand** stores so trips load instantly and stay readable offline; 65 **Jest** tests and a **GitHub Actions** pipeline (typecheck, lint, test → signed release APK on version tags).

LocalClaude — Design Board *Electron, React, TypeScript, Claude Agent SDK* *Open source · 2 merged PRs*

- Built the **Design** feature of an open-source Claude desktop app (~3.3k lines): describe a prototype, slide deck, wireframe or one-pager, and Claude lays it out as frames on an **infinite canvas** beside the chat — pan, pinch-zoom, marquee multi-select, present mode, version switching and click-to-comment revisions.
- Frames render live via a **streaming partial-JSON parser** that reads tool-call input mid-stream; multi-screen designs get a design system first, then fan out to up to 8 **parallel subagents**, one per screen.
- Added PNG (2×), PDF and HTML export (a whole deck as one PDF) rendered in a hidden, sandboxed window; sped up session startup by skipping MCP servers, Chrome and computer use; added unit and end-to-end tests.

Habex — Expense, Habit & Investment Tracker *React, TypeScript, Supabase (Postgres), shadcn/ui, Tailwind* *habex.santhu.site · GitHub*

- Live personal-finance app on my own domain: expense entry with recurring rules and monthly views, habit tracking with a 21-day challenge, shared investment groups, and an admin dashboard.
- Backed by **Supabase Postgres** with 12 versioned **SQL migrations** and three **edge functions** (PIN login, admin user management, activity audit) keeping privileged operations off the client.

FraudShield — Accounts-Payable Fraud Detection API *Python, FastAPI, SQLAlchemy, Pydantic*

- REST service that scores supplier invoices for fraud risk, with reviewer and admin workflows and auto-generated **OpenAPI** docs.
- Built a configurable **rules engine** — amount thresholds, per-vendor invoice velocity over a rolling window, and repeated round-amount detection — feeding a scoring service; paired with a **gradient-boosted** model trained on a labelled fraud dataset.

Time Tracker — Desktop Work-Hour Automation *Python, Tkinter, Selenium, PyInstaller*

- Desktop app that automates weekly timesheet entry: drives an attendance portal with **Selenium**, paging week by week and waiting on the rendered date before scraping each day's records rather than relying on fixed sleeps.
- Runs scraping off the UI thread and marshals results back to Tk's main loop, coalesces per-keystroke writes into one debounced save, and ships as a standalone Windows executable via **PyInstaller**.

EDUCATION

Siddaganga Institute of Technology — B.E. Electronics & Instrumentation, CGPA 8.94 **Dec 2020 – Jun 2024**

Vidyavahini PU College — Class 11 & 12, Pre-University (PCMB), 88.56% May 2018 – Jul 2020

Vivekananda Vidya Mandira School — Class X (SSLC), 78.64% May 2017 – May 2018

ADDITIONAL

Hack-o-Smart — 24-hour Tumkur Smart City hackathon; built a district-level education platform surfacing seat availability and fee details. ◆ **Certifications:** SQL (CodeChef), Digital Skills (HackerRank) ◆ **Leadership:** NCC Rank Holder, 4 Kar Tumkur Battalion